

Steve Chen

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EDUCATION

The Ohio State University

B.S. in Computer Science and Engineering / GPA: 3.53/4.00

Columbus, OH

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: Java, Python, C++17, C, SQL, TypeScript, JavaScript (ES6+), GDScript, Bash, HTML5/CSS3

Frameworks & Libraries: Spring Boot 3, FastAPI, Flask, React.js, Next.js, Redux, Node.js, JUnit, PyTest

Cloud, Data & Tools: AWS (Lambda, ECS, S3), Docker, PostgreSQL, Redis, Git, GitHub Actions, CI/CD, Linux

Systems & Practices: Distributed Systems, Data Structures & Algorithms, RESTful APIs, OAuth 2.0, WebSockets, Concurrency, Caching, Object-Oriented Design, Agile/Scrum, Unit & Regression Testing

EXPERIENCE

Python Software Foundation — Google Summer of Code 2026

Remote

*Open Source Software Engineer, **pocketpy** | Work Product*

May 2026 – Aug 2026

- Shipped the **Godot Vibe Plugin**, the project's flagship deliverable: prompt-driven asset generation, image editing, and scene automation inside the Godot editor (**GDScript** plugin + **FastAPI** backend).
- Eliminated arbitrary code execution by designing a structured-planning protocol in which the **LLM** fills a validated plan schema instead of emitting code, with a deterministic rule-based fallback when the model is unreachable.
- Unified **PixelLab** and **GPT Image** behind one provider interface and a five-stage pipeline, cutting a new art style from a code change to one **JSON** pack across **46 materials**, guarded by **6 regression test suites**.
- Extended **pocketpy**, a C-based Python interpreter, with standard-library internals and **typing.ReadOnly** support, maintaining the **prebuild.py** codegen that emits the **_generated.c** bindings.

K.C. Voicebotics (Conversational Voice AI)

Remote / Columbus, OH

Software Engineer Intern — Full Stack & Platform Integrations

Jan 2026 – Present

- Engineered a **multi-tenant OAuth 2.0** pipeline with dynamic redirect URIs and per-tenant token refresh, cutting new third-party integration onboarding from a code change to a configuration entry.
- Cut asynchronous JSON payload latency **25%** for **100+ concurrent clients** by restructuring **Spring Boot** routing and batching downstream API calls, exposed through a **React** operator console.
- Kept distributed third-party state consistent under failure with **idempotent** webhook handlers and exponential-backoff retries, powering near-real-time preview and approval workflows.

Intellicrane Tech (Industrial Automation)

Shenzhen, China

Software Engineer Intern — Backend & Computer Vision

May 2025 – Aug 2025

- Replaced manual assembly-line inspection with a hardware-in-the-loop computer vision service (**YOLOv5** + **Flask**), reaching a **95% defect detection rate** in production QA.
- Reduced edge inference latency **30%** through post-training **model quantization**, decoupled media ingestion into **AWS S3**, and covered REST and storage paths with **PyTest**.

LBASE Technology (E-Commerce Platform)

Shenzhen, China

Full Stack Software Engineer Intern

Jun 2024 – Sep 2024

- Cut API response latency **30%** under e-commerce traffic spikes by normalizing the **PostgreSQL** schema, adding covering indexes, and introducing a **Redis** cache-aside layer.
- Eliminated duplicate charges on client retries by hardening **Stripe** payment flows with **idempotency keys** and transactional writes, gated by **GitHub Actions** CI ahead of **Docker** deploys.

PROJECTS

SpellTide Godot Native Extension | C++17, CMake, godot-cpp, MessagePack, KCP

Mar 2026 – Present

- Built the cube-body physics layer of a 2D torus-world game in **C++17**: chunked spatial partitioning for broad phase, axis-aligned face contacts with penetration separation for narrow phase, removing resting-body jitter.
- Held frame time flat as entity count scaled by batching tile draws through **MultiMesh** instancing and deriving a coverage formula that rasterizes circular bodies onto a square tile grid.
- Implemented **MessagePack** serialization for save packing and lockstep **multiplayer** sync over a **WebSocket** control channel and a **UDP/KCP** data channel.

Serverless AI Support Agent | Java 21, Spring Boot 3, React, AWS Lambda/ECS, PgVector

Dec 2025 – Present

- Architected an autonomous support agent that resolves multi-step requests by invoking external APIs through **LLM tool calling** over **MCP**, grounded in a **PgVector** RAG retrieval layer.
- Drove idle compute toward zero with stateless **Java 21** handlers on **AWS Lambda**, using **ECS** for long-running jobs.